

Misinformation Detection Using Transformer Language Models

P.1: Project Topic Description

AI and Natural Language Processing

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This project addresses the problem of multi-class veracity classification — predicting whether a political or news statement is *true*, *mostly-true*, *barely-true*, *half-true*, *false*, or *pants-on-fire* — using fine-tuned pre-trained transformer language models applied to the LIAR benchmark dataset. This topic was chosen because disinformation and fake news have well-documented harmful effects on democratic processes, public health, and social cohesion, yet manual fact-checking pipelines cannot operate at the speed and scale demanded by modern digital media; NLP-driven automated classification therefore represents both a pressing real-world need and a technically demanding research problem involving multi-class label imbalance, short-text contextual representation, and the fusion of unstructured text with structured speaker metadata. Wang (2017) introduced the LIAR dataset — 12,836 labeled political statements from PolitiFact.com pre-split into training, validation, and test sets — and demonstrated that logistic regression, support vector machine, and LSTM baselines achieve only modest accuracy on the six-class task (approximately 27%), establishing the benchmark this project targets for improvement; the paper also showed that incorporating structured speaker metadata (party affiliation, credit-history vector of prior label counts) provides complementary signal beyond statement text alone, directly motivating the metadata-fusion approach planned here. Devlin et al. (2019) presented BERT, the bidirectional transformer pre-training paradigm whose deep contextual representations encode subtle linguistic patterns that distinguish shades of truth from fabrication, and which serves as the primary model architecture to be fine-tuned in this project; BERT’s ability to capture long-range dependencies makes it markedly better suited to granular veracity classification than shallow feature-engineered approaches. Shu et al. (2017) systematically surveyed fake news detection methods across linguistic, social-network, and user-level feature spaces and demonstrated that purely text-based classifiers plateau at moderate accuracy — a structural limitation that motivates the plan to augment BERT’s text representations with LIAR’s speaker features through a multi-input feedforward fusion layer in a subsequent experimental phase of this project.

References

1. Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Volume 1, pp. 4171–4186. Minneapolis, MN: Association for Computational Linguistics. <https://doi.org/10.18653/v1/N19-1423>.
2. Shu, K., Sliva, A., Wang, S., Tang, J., and Liu, H. 2017. Fake News Detection on Social Media: A Data Mining Perspective. *ACM SIGKDD Explorations Newsletter* 19(1): 22–36. <https://doi.org/10.1145/3137597.3137600>.
3. Wang, W. Y. 2017. “Liar, Liar Pants on Fire”: A New Benchmark Dataset for Fake News Detection. *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*, Volume 2: Short Papers, pp. 422–426. Vancouver, Canada: Association for Computational Linguistics. <https://doi.org/10.18653/v1/P17-2067>.